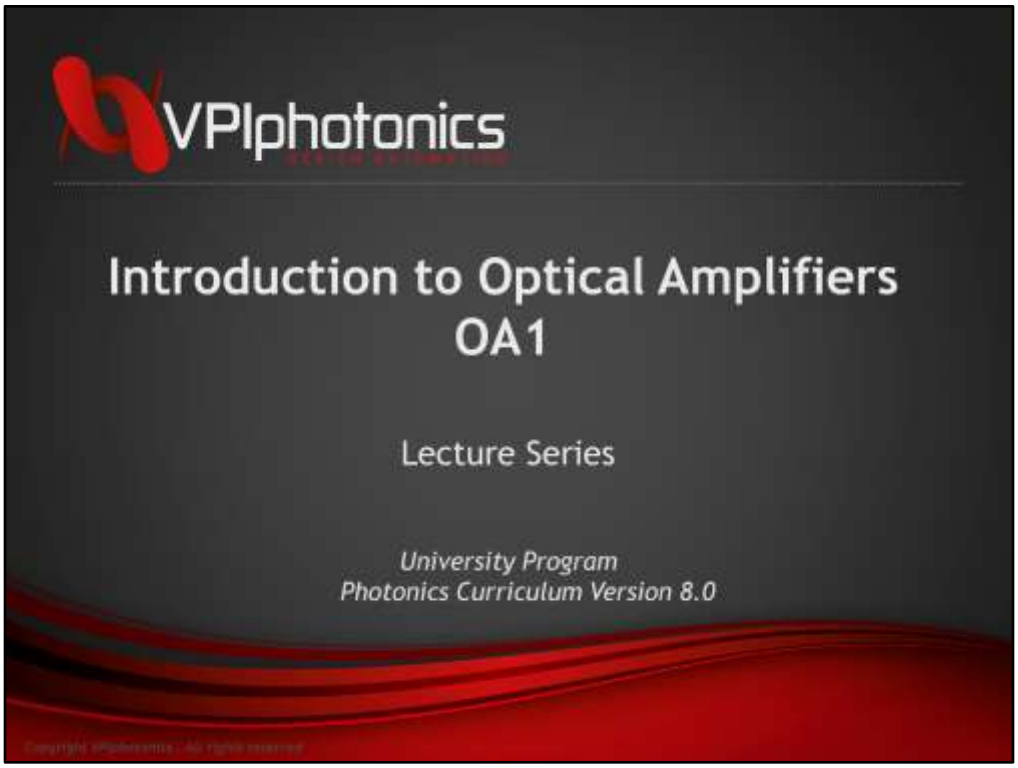
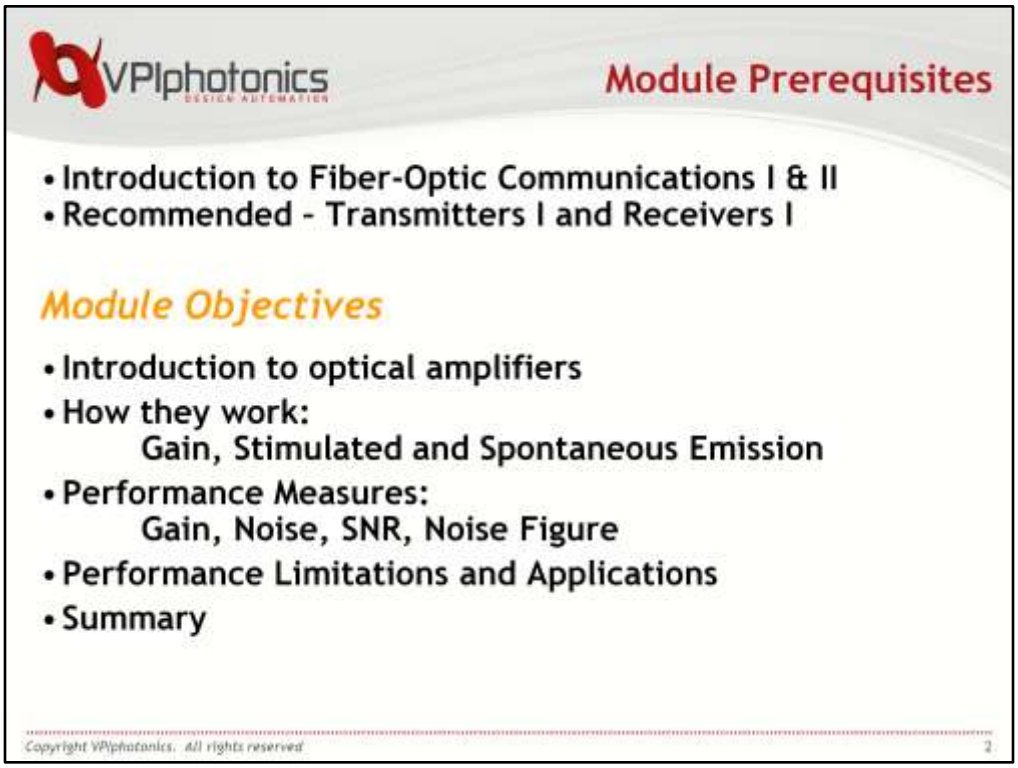
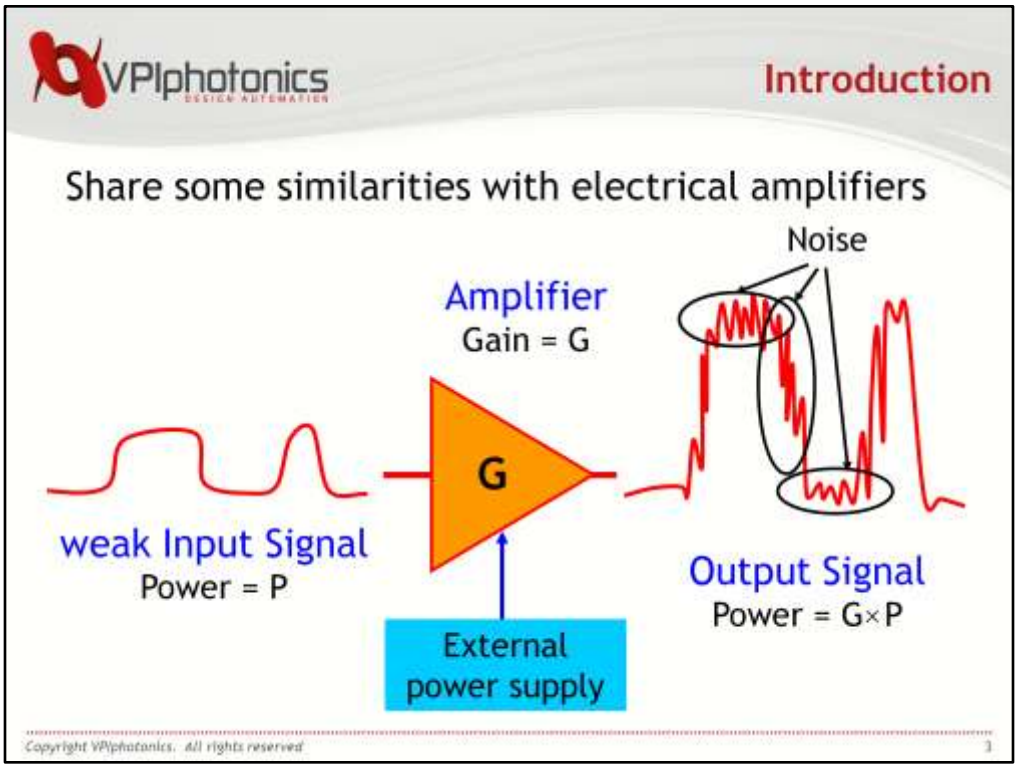
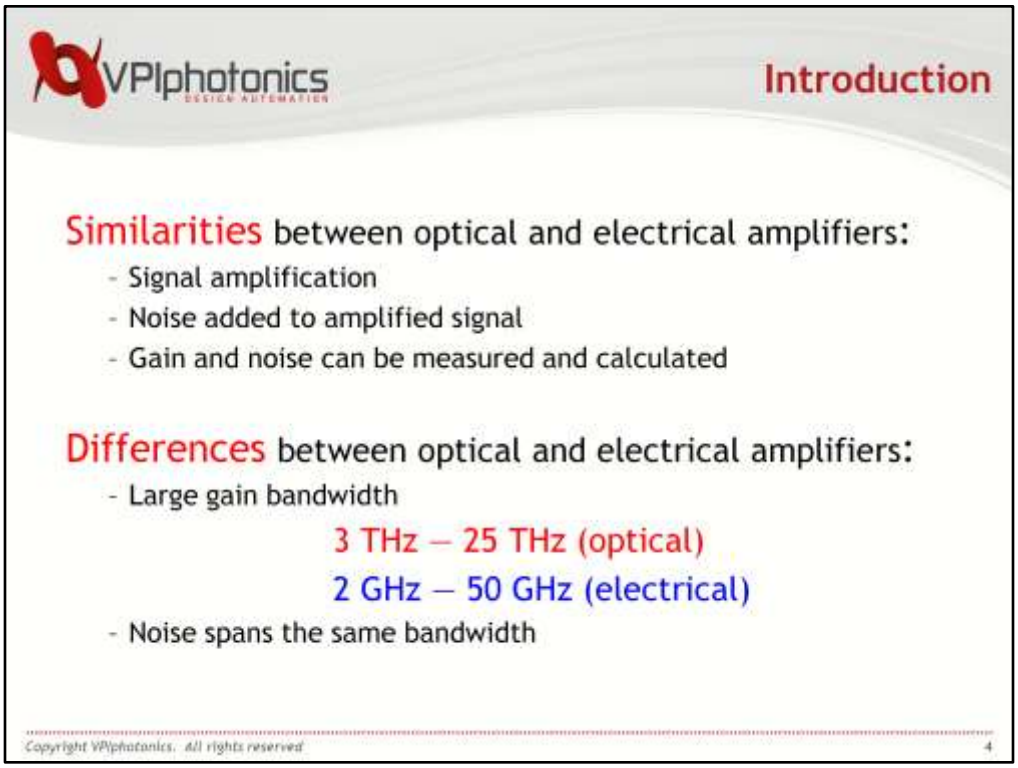

VPIphotonics
Introduction to Optical Amplifiers
OA1
Lecture Series
University Program
Photonics Curriculum Version 8.0



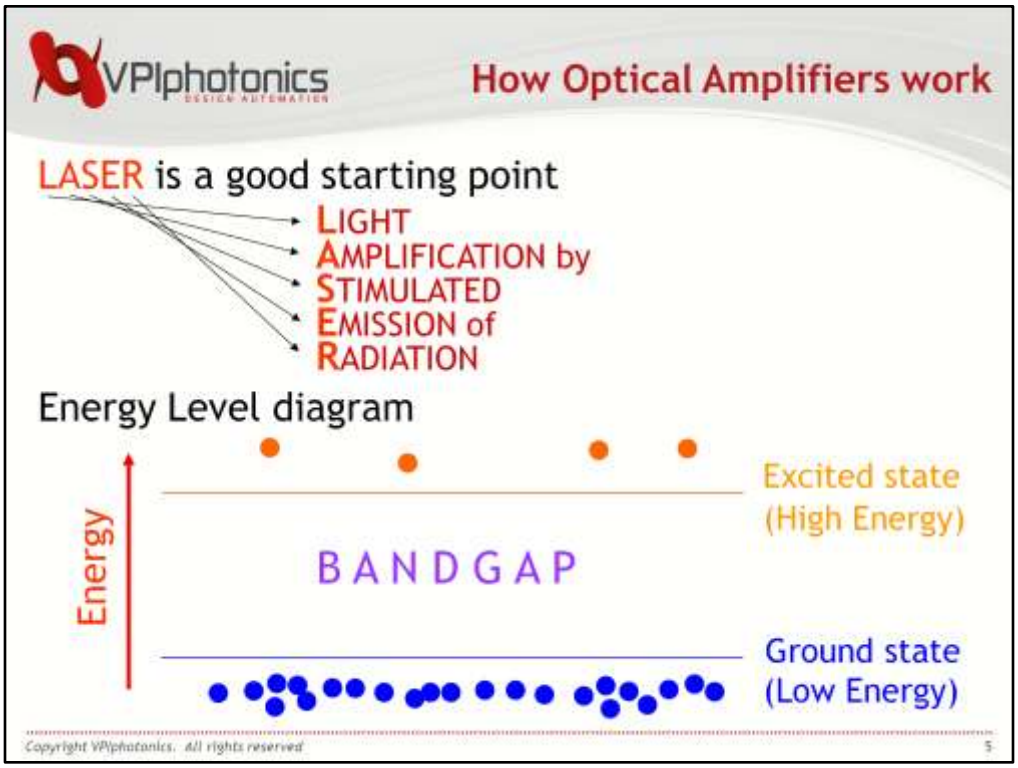
Module prerequisites and objectives – edit to suit the specific requirements of your course.



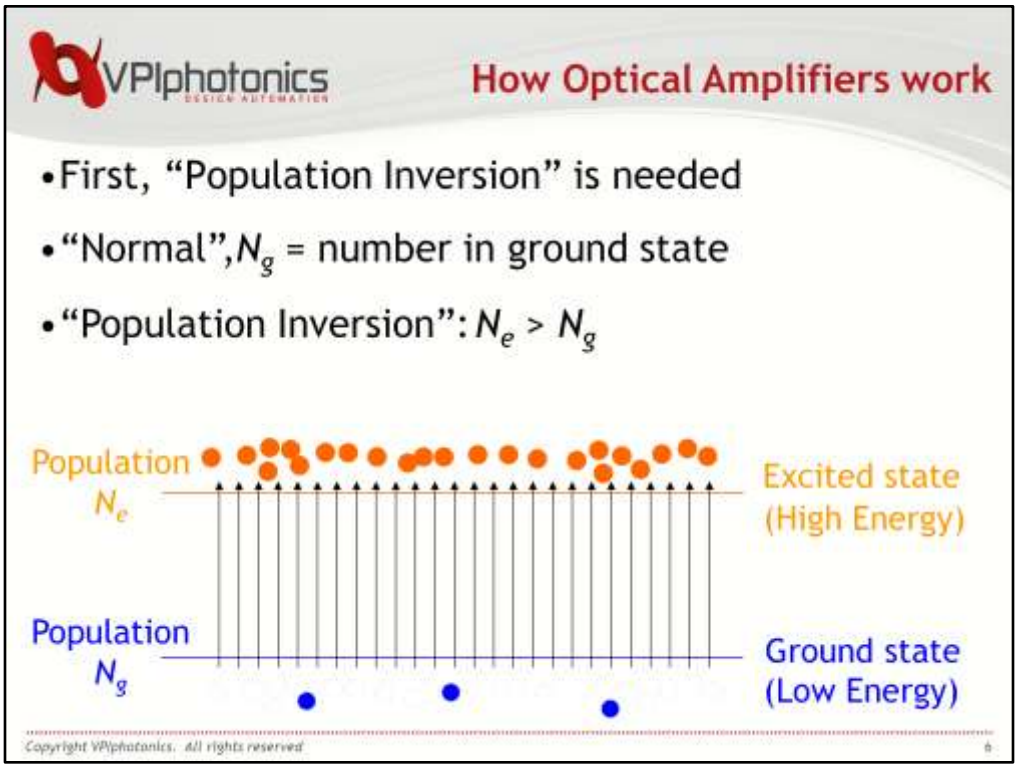
In the earliest days of what came to be called electronics, ways of amplifying weak electrical signal did not exist…"(Green, p. 219)
Someday someone will say the same thing, replacing "electronics" with "photonics", and "electrical" with "optical"
The earliest days for optical amplification will be the 1980's
At the highest level of description, optical amplifiers are similar to electrical amplifiers.
Noise is typically represented by an equivalent source at the input or output, based on measurement or calculation of the noise power spectral density.



- A weak input signal can be amplified to a strong output signal
- The output signal is accompanied by broadband noise from the amplifier.
- Gain and noise can be measured and calculated.



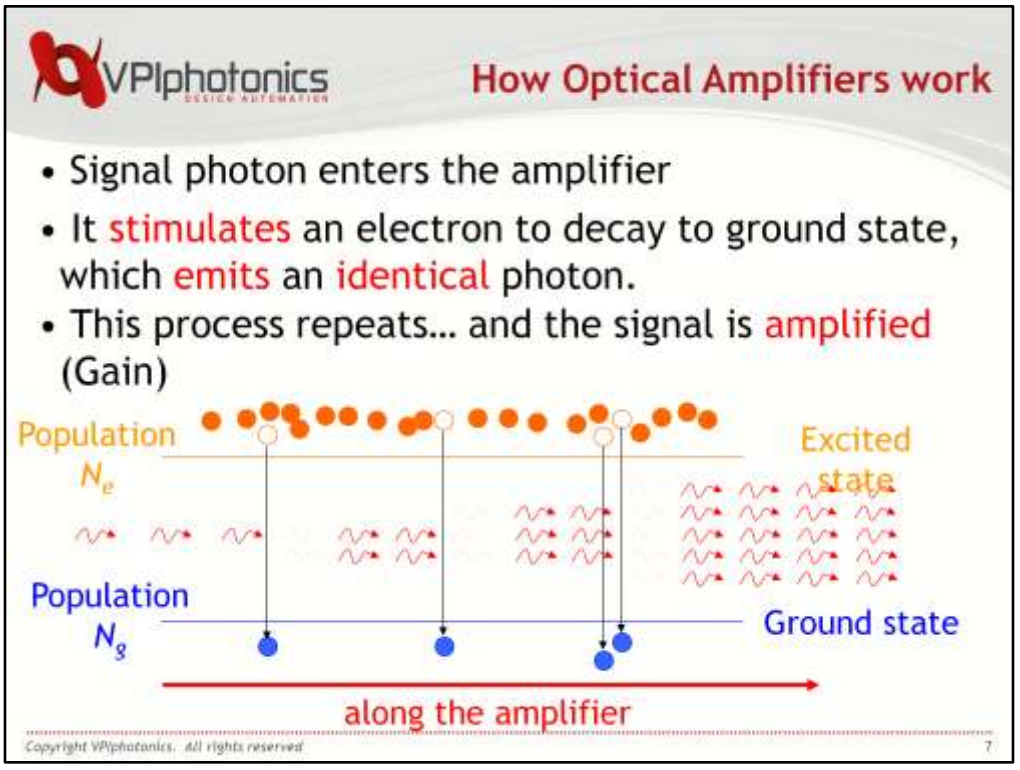
• An optical amplifier can be built based on any structure that lases (produces laser light).
• After all, LASER = Light **Amplification** by Stimulated Emission of Radiation
• This will be explained with the help of an Energy Level diagram (simplified to a two-level diagram, of course, for the time being).
• Here is one such diagram for the material in an optical amplifier that will do the amplifying
• The electrons in the material are normally in the low energy state (blue dots)



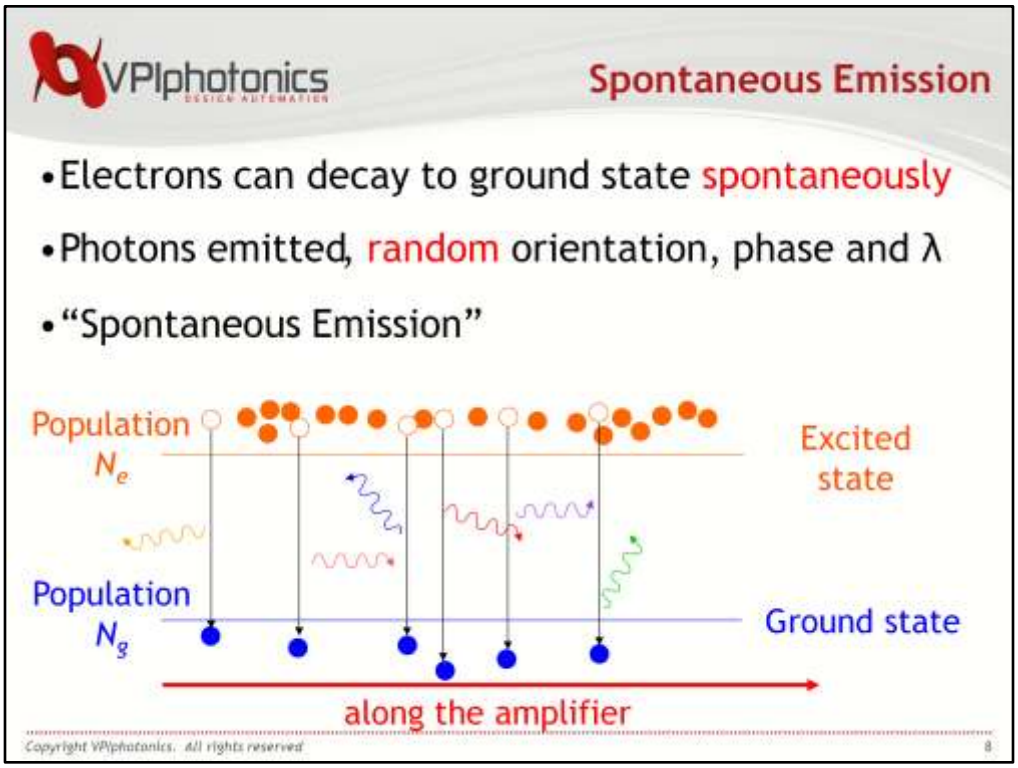
• For amplification to occur, at first "Population Inversion" is needed in the material.
• What does "Population Inversion" mean? Well normally, the majority of the molecules of the material is in the low energy state.
• When Population Inversion occurs, the majority of the atoms of the material are in the high energy state.
• Question: How does Population Inversion occur?
• Well, we need an external source of energy/power to make this happen.

(Potential additional explanatory slides)

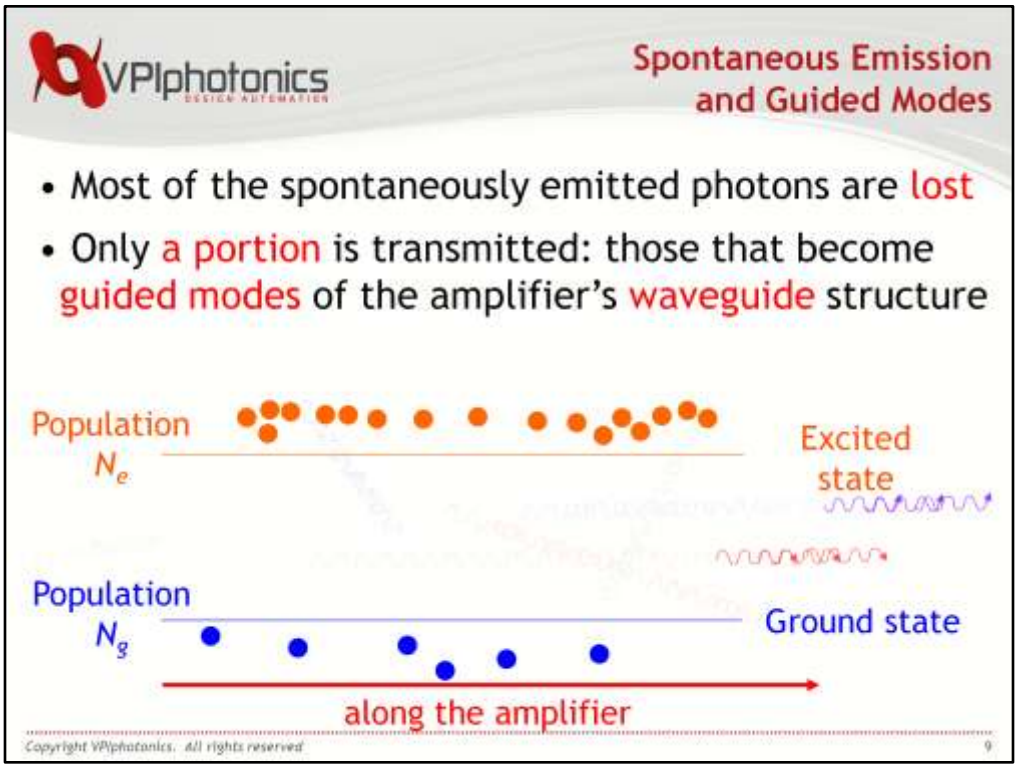
• Like Electrical Amplifiers, this external source can be electrical (e.g. power for amplification comes from the electrical power supply rails). This is called "electrical pumping" – we "pump" up the energy of the atoms to make them undergo population inversion.
• The external source can also be optical. This is called "optical pumping". We "pump" up the energy of the atoms by illuminating the material with an external light source!



• Once population inversion is achieved, the material is able to amplify a signal.
• Here's how: A signal photon enters the excited material.
• It stimulates one of the excited electrons to return to the ground state, which emits a photon identical to the signal photon.
• This can occur over and over again as the signal photon(s) pass(es) through the material, thus amplifying the signal.



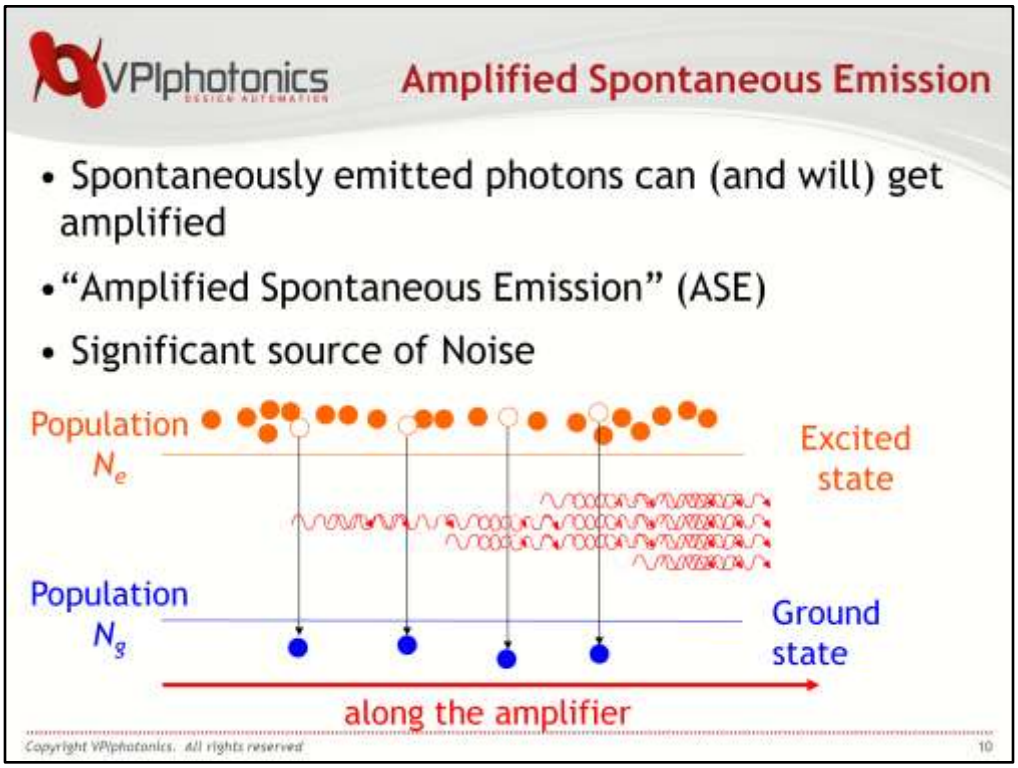
• Now, excited atoms can also decay back to the ground state spontaneously (without being stimulated by any signal photon).
• When they decay spontaneously, photons are also emitted, but these have random phase and orientation.
• This is known as "Spontaneous Emission", and is the origin of noise in optical amplifiers.



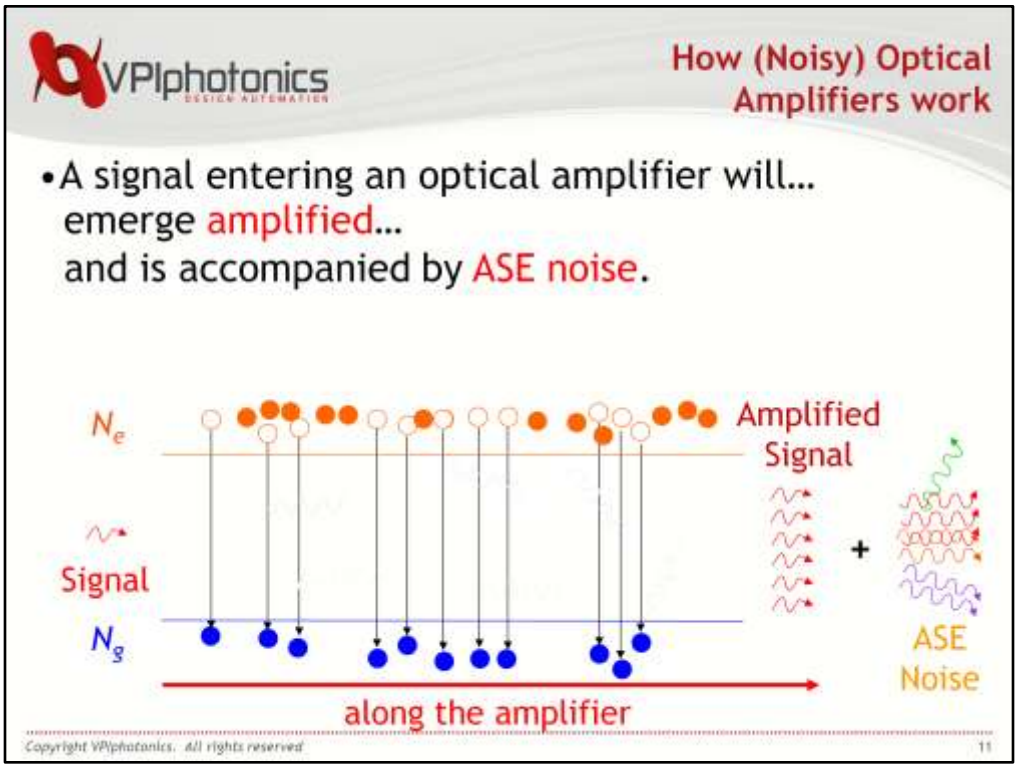
• Most of the spontaneously emitted photons are lost, because they do not couple into the guided modes of the waveguide structure of the amplifier.

• As such, only a portion of these spontaneously emitted photons are transmitted – those that couple into the guided modes of the amplifier.

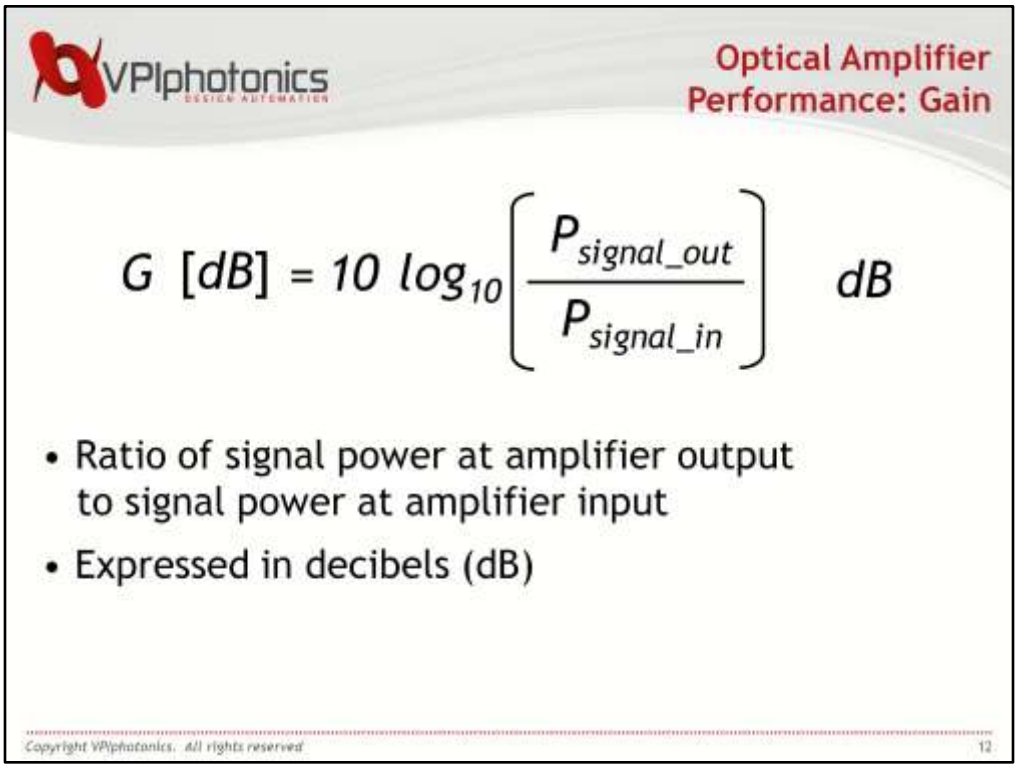
• Signal photons are already propagating as a guided mode of the amplifier waveguide structure, so the emitted photons that they stimulate will also propagate in the same guided mode.



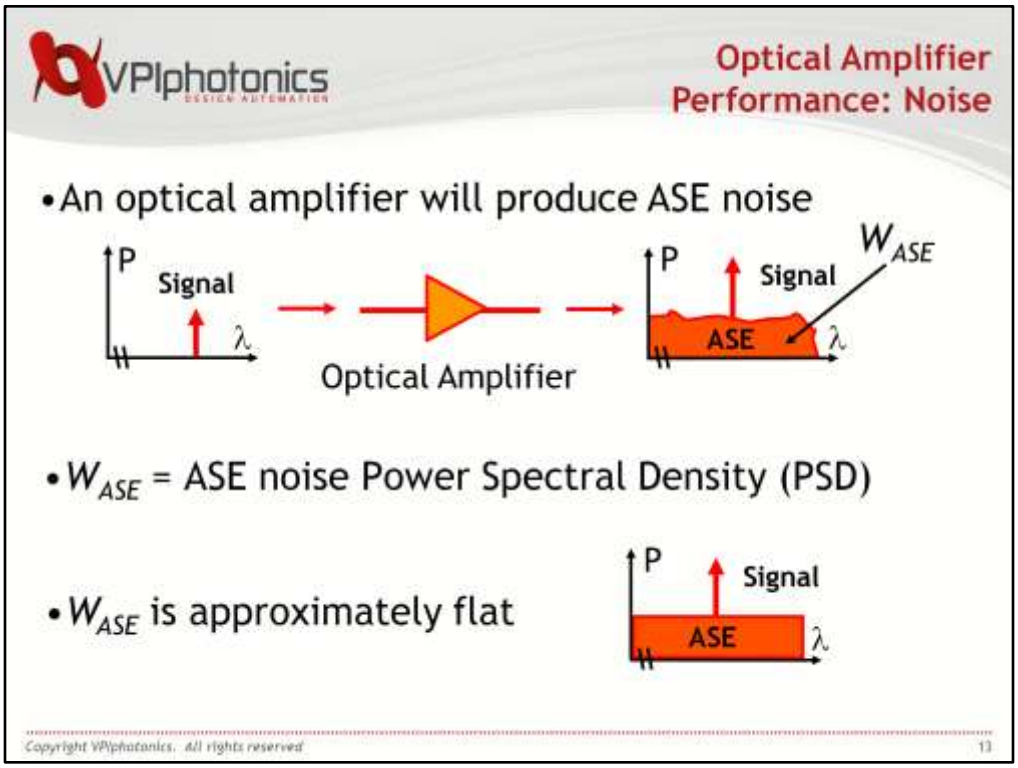
• Spontaneously emitted photons traveling through the excited material of the optical amplifier will also be amplified.
• This is known as "Amplified Spontaneous Emission (ASE)", and is a significant source of optical noise.
• This source of noise is also termed "ASE noise".



A signal entering an optical amplifier will emerge amplified with added ASE noise.



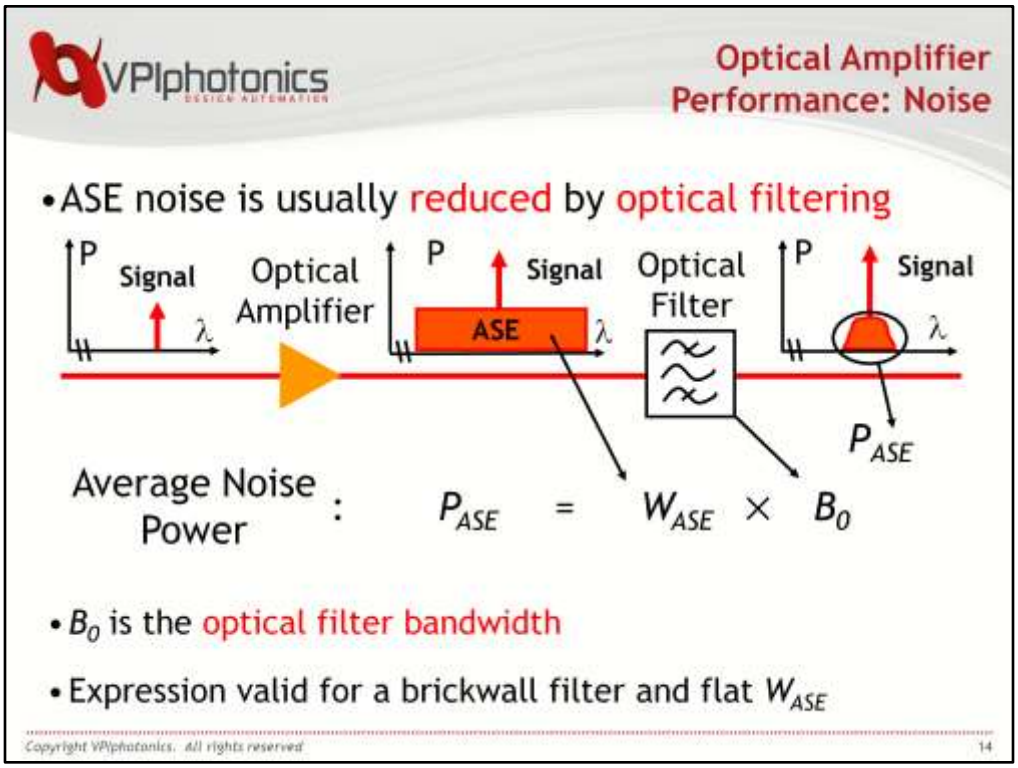
G, the gain of an amplifier, is the ratio of the output signal power to the input signal power and is customarily expressed in decibels (dB).



• An optical amplifier, besides amplifying an signal, also produces Amplified Spontaneous Emission noise.

• In a practical amplifier, the ASE noise extends over the entire gain bandwidth and may be very wide band.

•The ASE noise power spectral density (PSD) is approximately flat (over the bandwidth of the amplifier) .



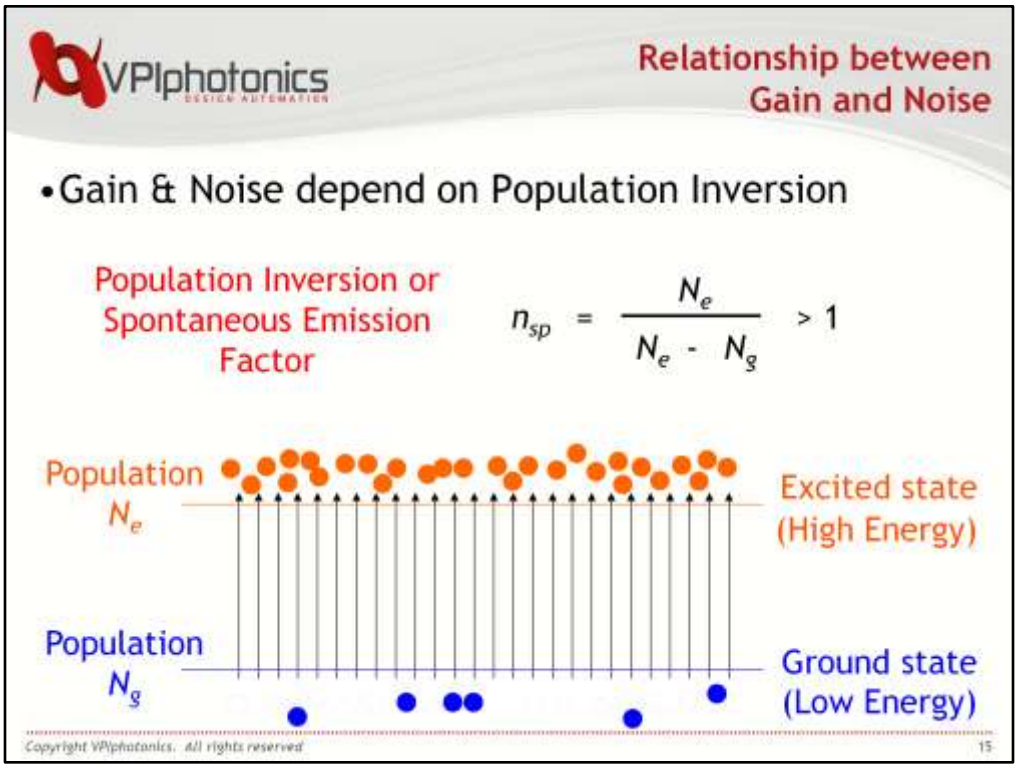
• In real systems, an optical bandpass filter will be used, either immediately after the optical amplifier, or just in front of the receiver, to eliminate most of the ASE noise.

•Thus, the average (optical) ASE noise power *(P_{ASE})* that will be detected after an optical filter of bandwidth B_0 is:

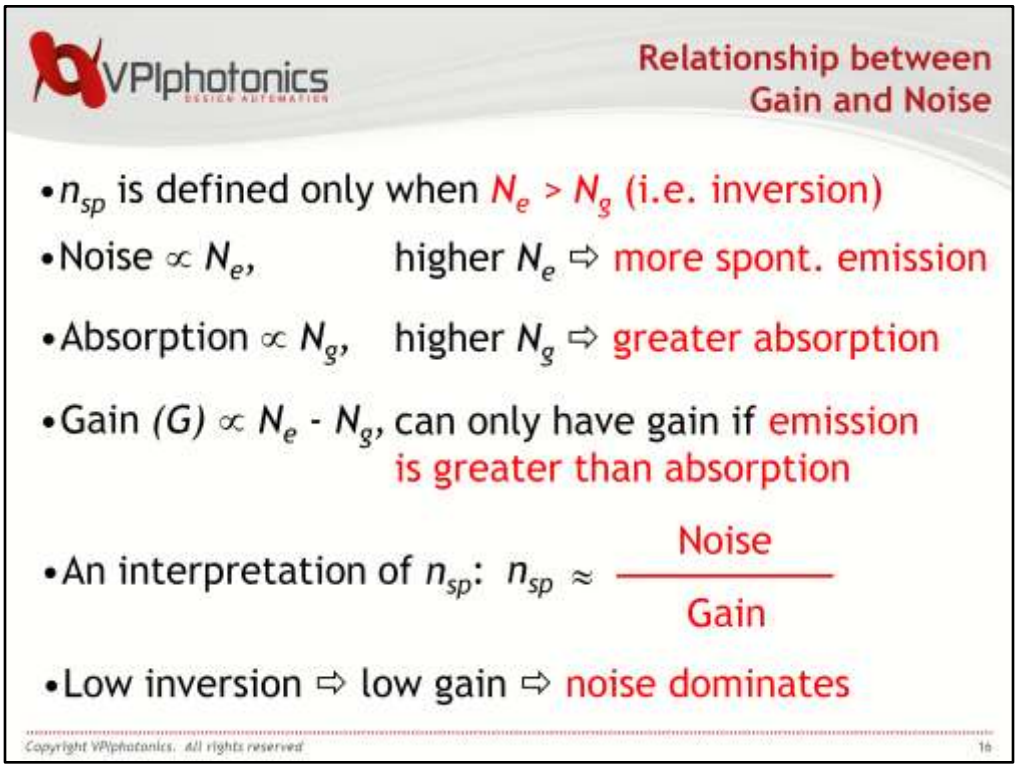
$$P_{ASE} = W_{ASE} B_0$$

• where W_{ASE} is the output power spectral density (per polarization state) expressed in W/Hz.

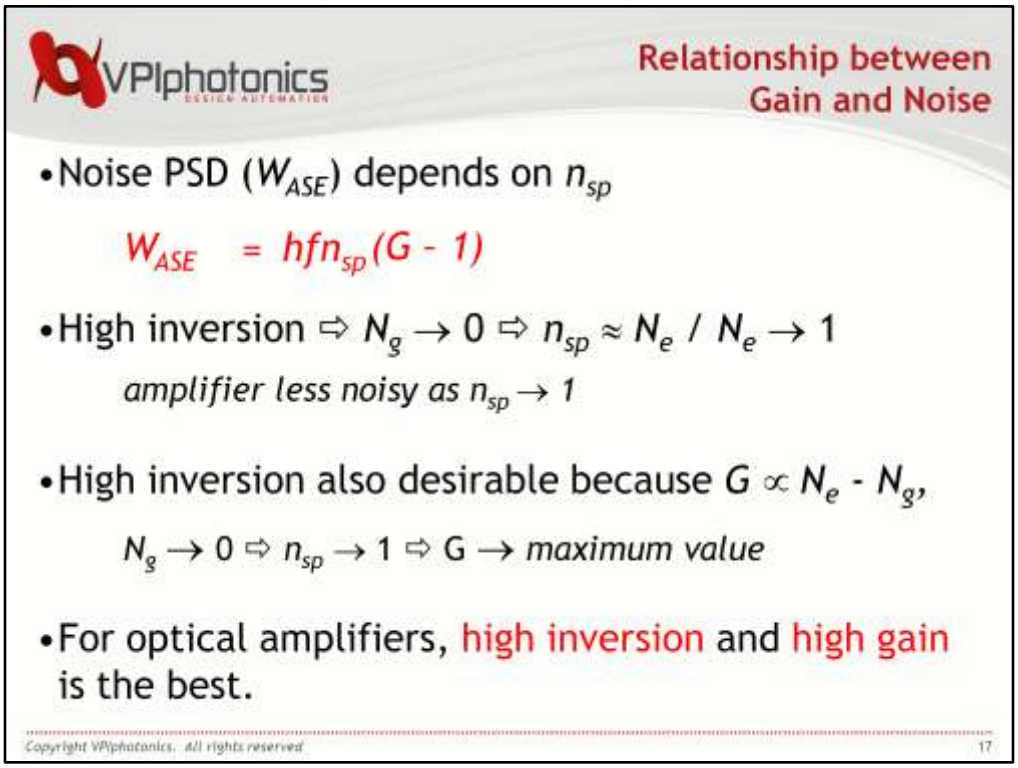
•This expression above assumes a flat ASE noise PSD and a brickwall filter of bandwidth B_0. For a non-flat PSD and/or realistic filter function, we will have to integrate over the bandwidth of the filter to obtain an accurate expression for P_{ASE}.



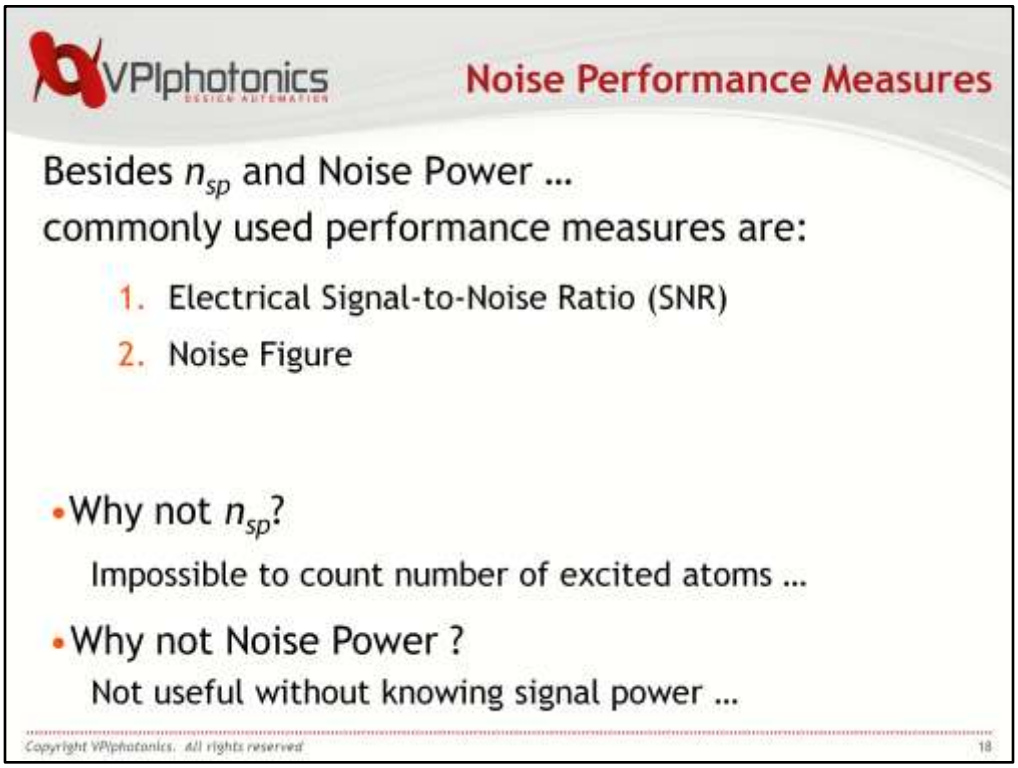
• Both gain and noise power depend on population inversion (i.e. how many more atoms are in the excited state than in the ground state).
• The exact relationship is determined by the Einstein equations (in thermal equilibrium).
• The Population Inversion factor (also known as the Spontaneous Emission factor, which is why the "sp" appears as a subscript in n_{sp}) is used to describe the extent of population inversion.
• Note that $n_{sp} > 1$.
• Question that may be asked: Under normal circumstances, $N_g >> N_e$, n_{sp} is negative! So how do we reconcile this with our assertion that $n_{sp} > 1$?



• Note that this expression for n_{sp} is valid only when there is population inversion (i.e. $N_e > N_g$, the material is providing amplification or gain). This is the answer to the question in the notes of the previous slide.
• Noise is proportional to the excited state population. More atoms in the excited state means higher probability for them to spontaneously decay and emit photons.
• Absorption of photons is proportional to the ground state population. More atoms in the ground state means higher probability of photons being absorbed.



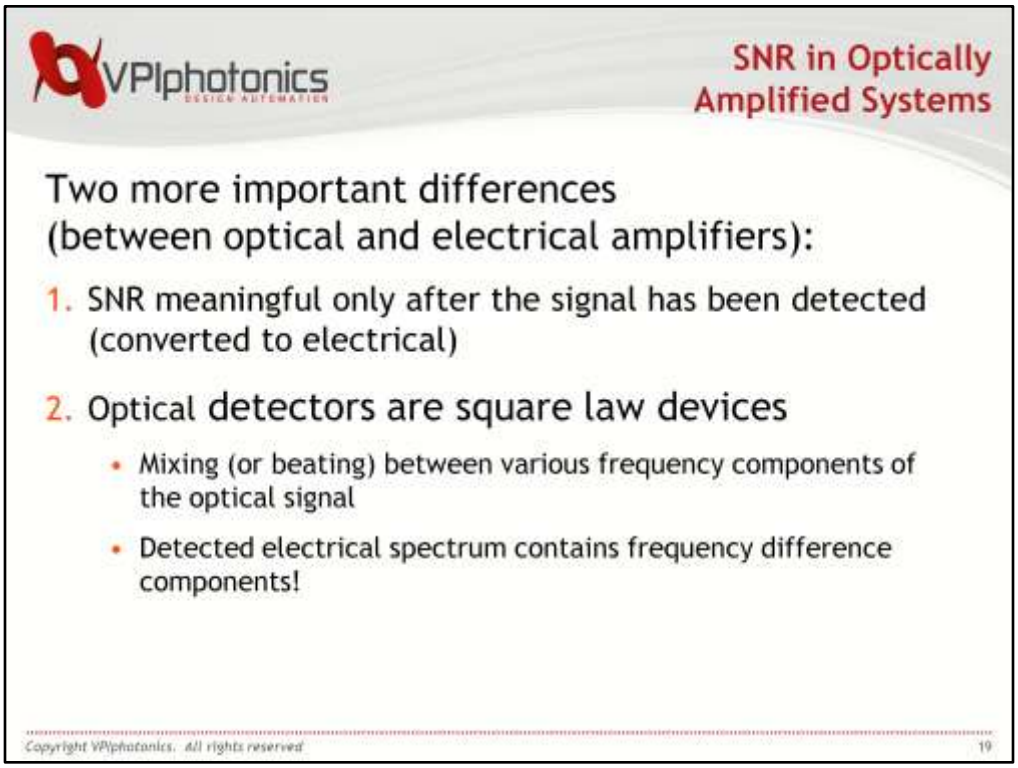
- Both Gain and Noise power depend on population inversion (n_{sp}) . h = Planck's constant.
- Noise PSD decreases as n_{sp} decreases towards the theoretical minimum value of 1
- Gain increases as n_{sp} decreases towards the value of 1, reaching a maximum when n_{sp} = 1.



Besides n_{sp} and noise power, commonly used performance measures are the electrical SNR after the photodetection of the optical signal, and the noise figure of the amplifier (noise figure will be defined later).

n_{sp} is difficult to obtain directly (impossible to count the number of excited atoms) and difficult to measure or obtain indirectly, whereas SNR is a well established measure of the noise performance of (electrical) amplifiers (and it is desirable to find similar measures for photonic devices). While the noise power is easy to measure, it is not useful without knowing what the signal power is, and SNR provides the comparison of the noise power to the signal power.

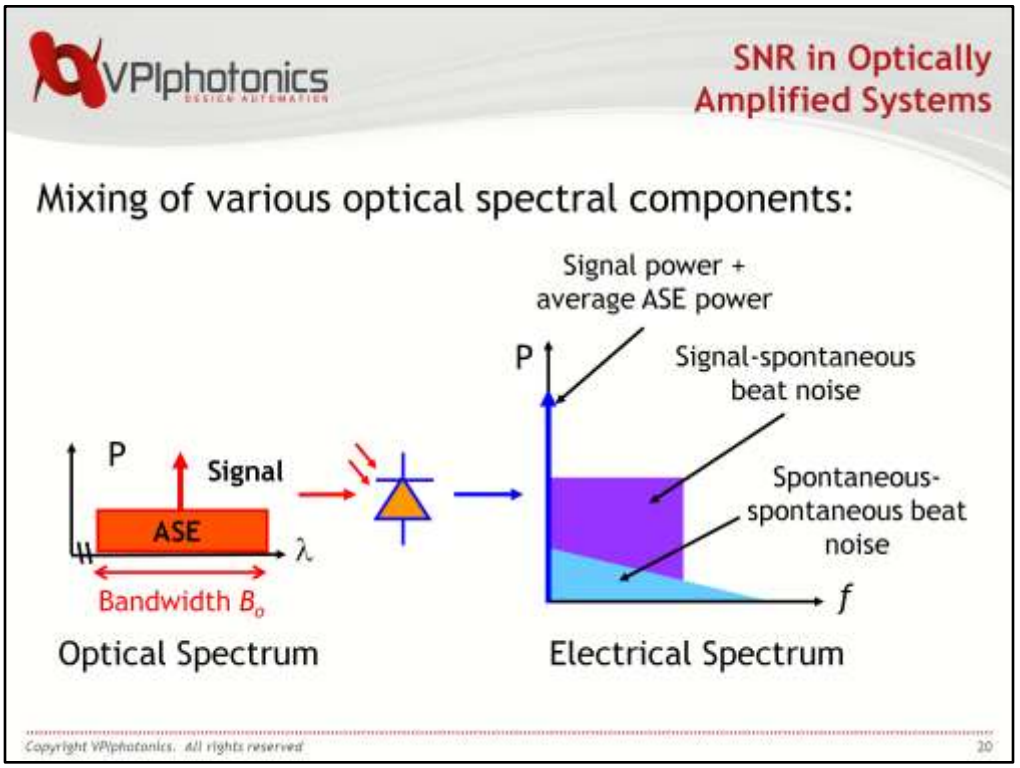
The NF of a device is defined as the ratio of the SNR at the input of the device to the SNR at the output of the device. This is a practical performance measure as the SNR at the input and output of a device is easy to measure. In datasheets, the noise performance of a device is often given as a NF. This noise performance measure (and how it will apply to optical amplifiers) will be discussed in later slides.



Two more important differences between optically amplified and electrically amplified systems that students should appreciate:

1) It is meaningless to consider SNR of the optical system BEFORE the optical signal is detected by the receiver (and converted to an electrical signal)
2) More crucially, optical detectors are square law devices. A square law detector will cause mixing between the different frequency components of the optical signal. This results in a detected electrical spectrum that contains additional "mixing" components that correspond to the frequency differences between the spectral components of the optical field!

These differences make the calculation of the SNR of an optically amplified system slightly more involved (than for electrically amplified systems).



Mixing of two frequency components in a square law device can be explained using simple mathematics:

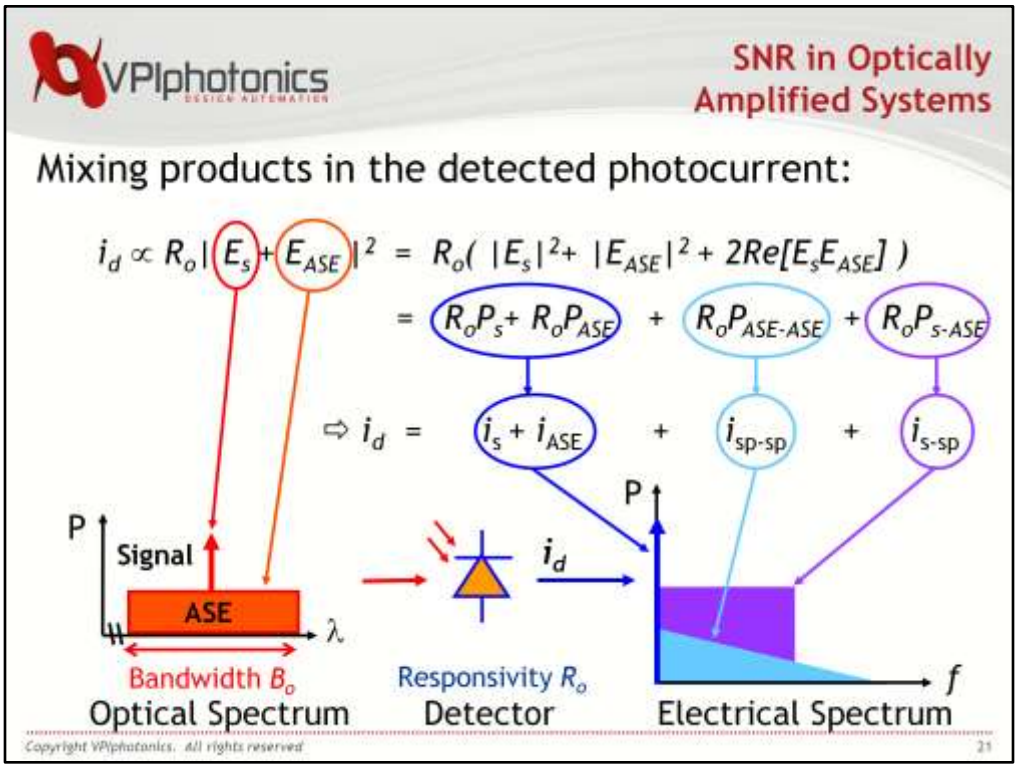
$[\cos(\omega_1 t) + \cos(w_2 t)]^2 = \cos^2(w_1 t) + \cos^2(w_2 t) + 2\cos(w_1 t)\cos(w_2 t)$

The third term on the right hand side of the expression above can be replaced (by trigonometric addition theorems):

$2\cos(w_1 t)\cos(w_2 t) = \cos[(w_1 + w_2)t] + \cos[(w_1 - w_2)t]$

The terms of the right hand side of the trigonometric expression above are the mixing terms.

Essentially, mixing occurs between the signal and the spontaneous emission noise, and the spontaneous emission noise with itself. The mixing between signal & spontaneous emission gives the rectangular distribution in the electrical spectrum, while mixing between spontaneous emission & spontaneous emission gives the triangular distribution.



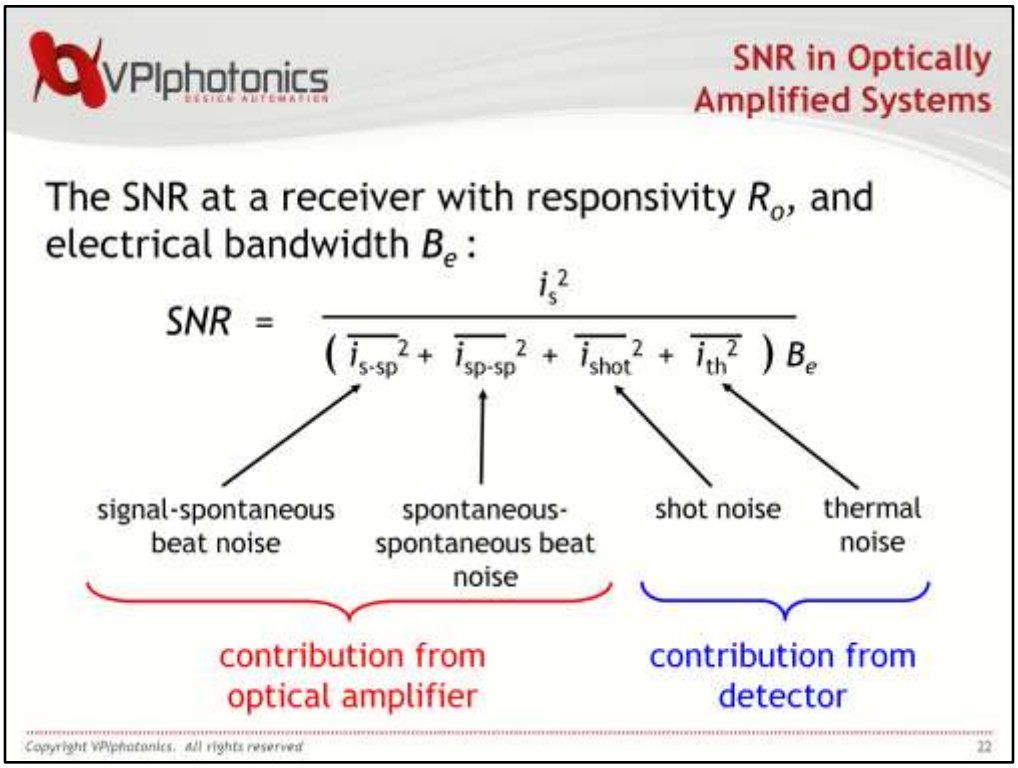
The detected current i_d is proportional to the square of the (electric field of the signal + ASE). The squaring gives rise to the various components of the electrical spectrum.

P_s = signal power

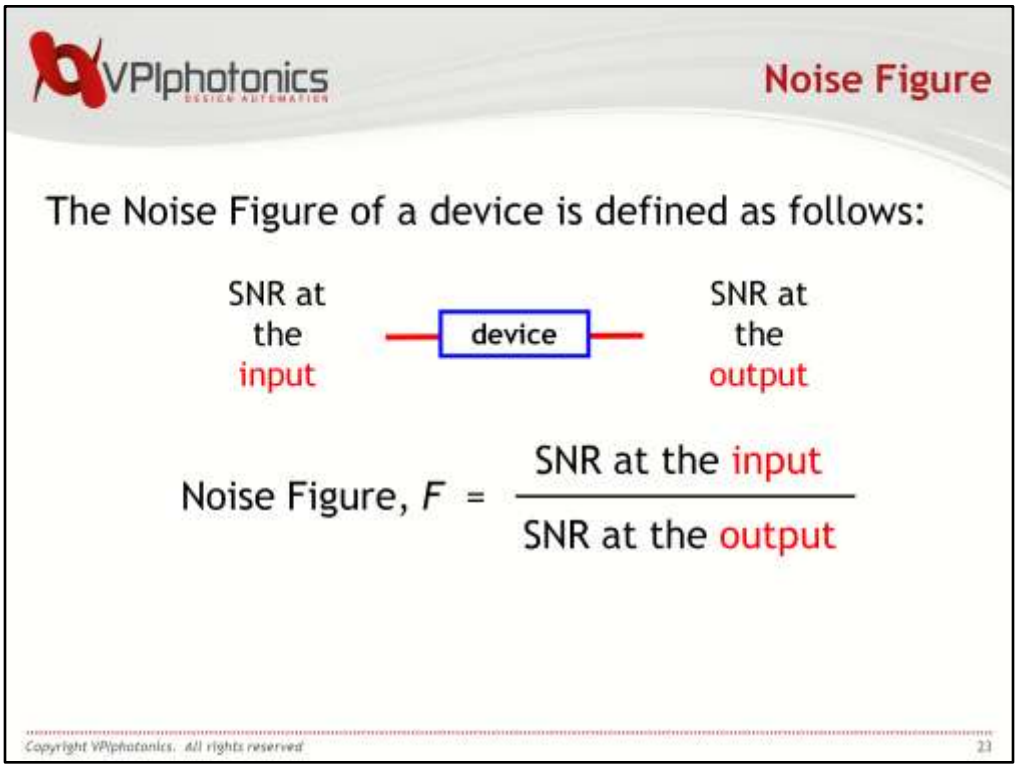
P_{ASE} R_0 B_0 =

= time varying

= time varying



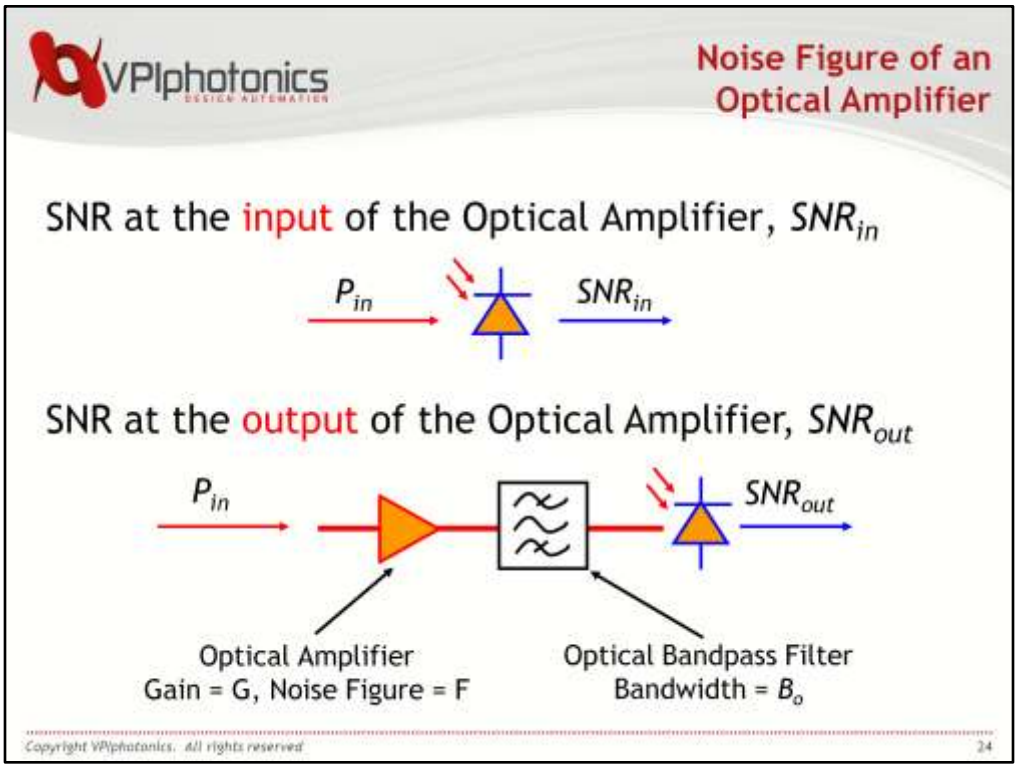
The SNR of the received signal can be written as shown in the slide. Note that the two terms in the denominator, the Shot noise and Thermal noise terms, are contributions from the photodetector. The details of photodetector/receiver noise will be covered in a separate module.



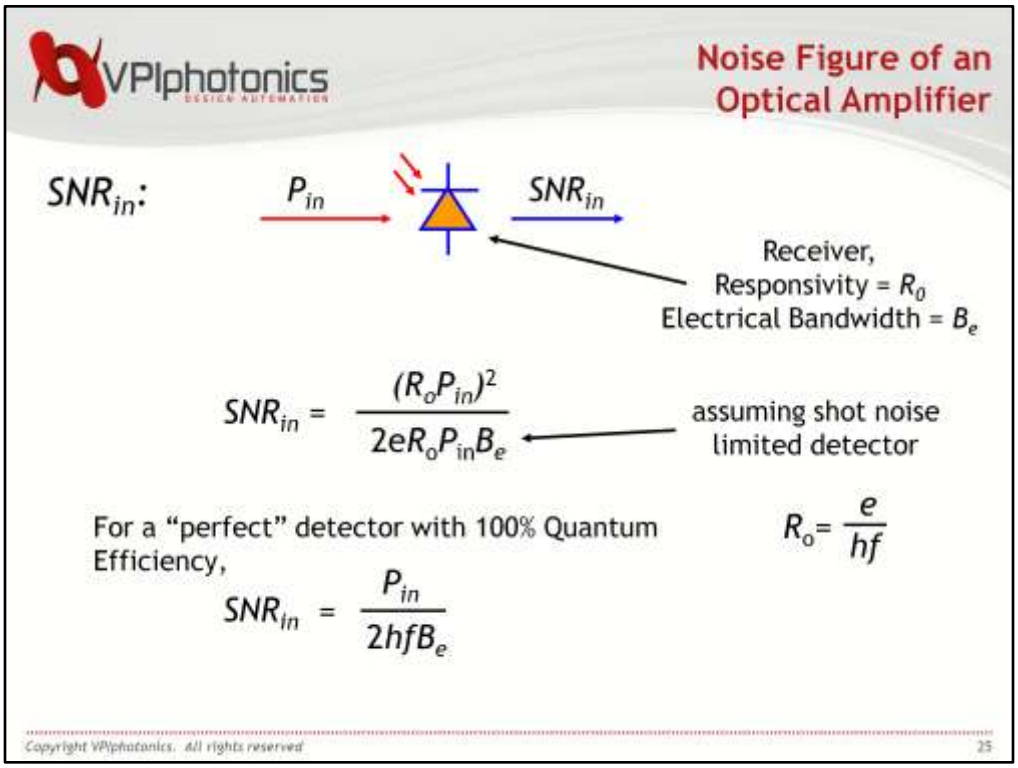
This slide defines the Noise Figure *(F)* of a device.

The noise figure of an (electrical) amplifier is defined as the ratio of the SNR at the input of the amplifier to the SNR at the output of the amplifier (i.e. SNR degradation), under certain standard conditions (source/load resistance, temperature).

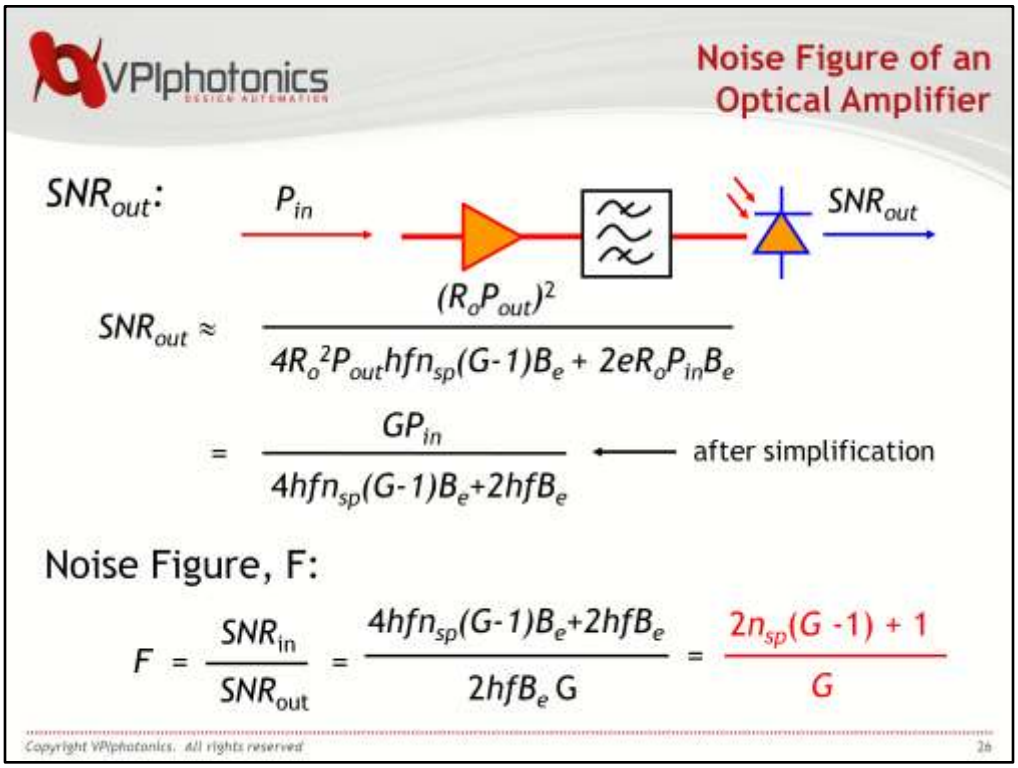
Similarly, the noise figure of an optical amplifier is the SNR degradation caused to an "ideal" input signal (shot noise only), after passing through the amplifier, as measured using an imaginary "perfect" detector (with no thermal noise and 100% quantum efficiency).



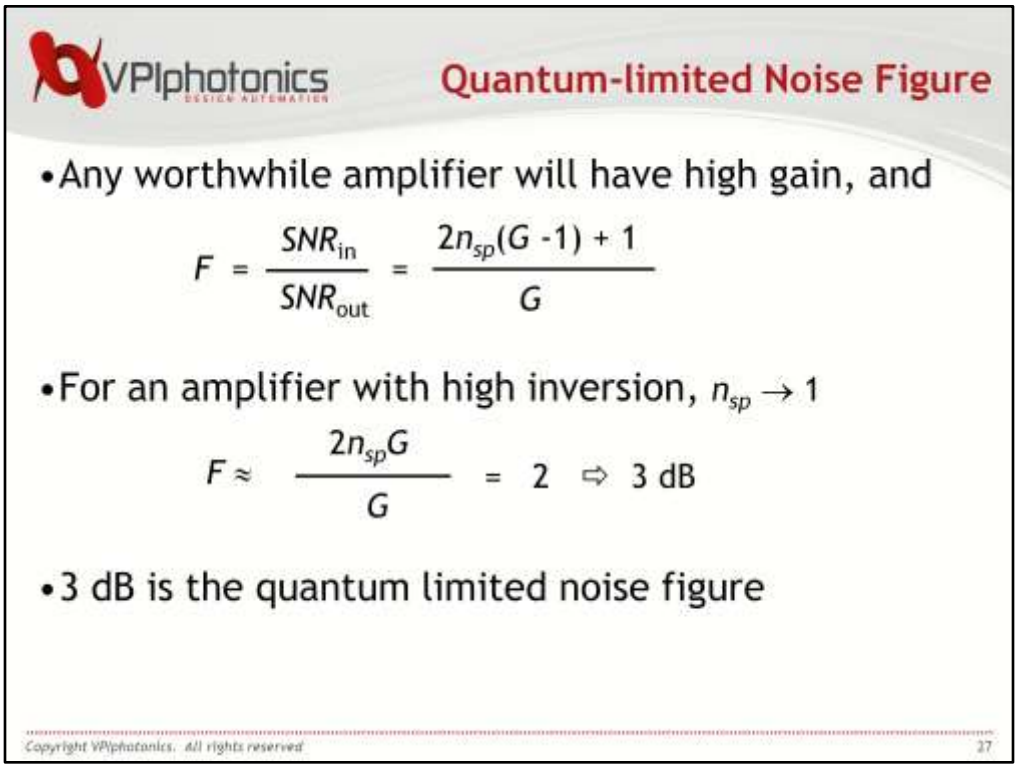
Here it is shown how the noise figure for an optical amplifier is measured. The electrical SNR at the input (SNR_{in}) is obtained by measuring with a photodetector at the input. The electrical SNR at the output (SNR_{out}) is similarly measured, but note that the measurement includes the effect of an optical bandpass filter (of bandwidth B_0) that is commonly used to reduce ASE noise at an amplifier's output.



The noise figure of an optical amplifier is written in terms of parameters of the photodetector (responsivity and electrical bandwidth) and input power.



The derivation of an expression for noise figure in terms of the gain (G) and spontaneous emission factor(n_{sp}) of the optical amplifier.

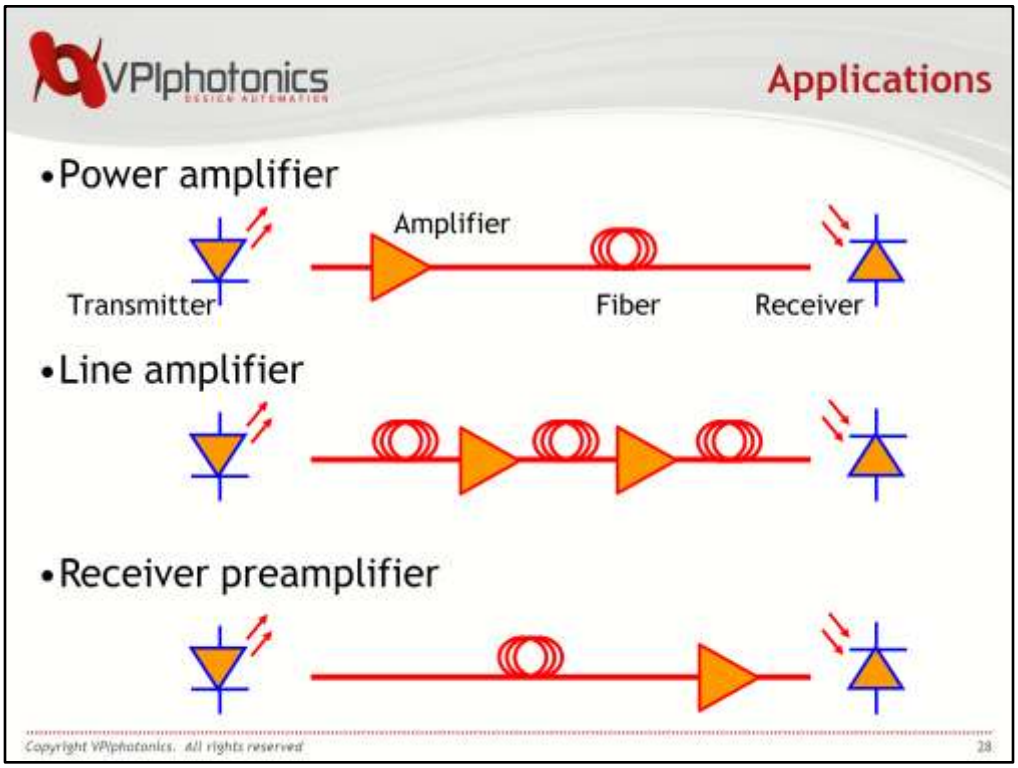


The expression for noise figure derived in previous slides enable some conclusions to be drawn. Firstly, most useful amplifiers will have high gain, so the noise figure expression can

be simplified further to: F $2n_{sp}G/G = 2n_{sp}$.

Theoretically, the minimum possible spontaneous emission factor (n_{sp}) is 1, so the minimum possible noise figure is 2, or 3 dB. This is a fundamental quantum-limit on amplifier noise performance, i.e. it is not possible to build a practical optical amplifier with a noise figure less than 3 dB.

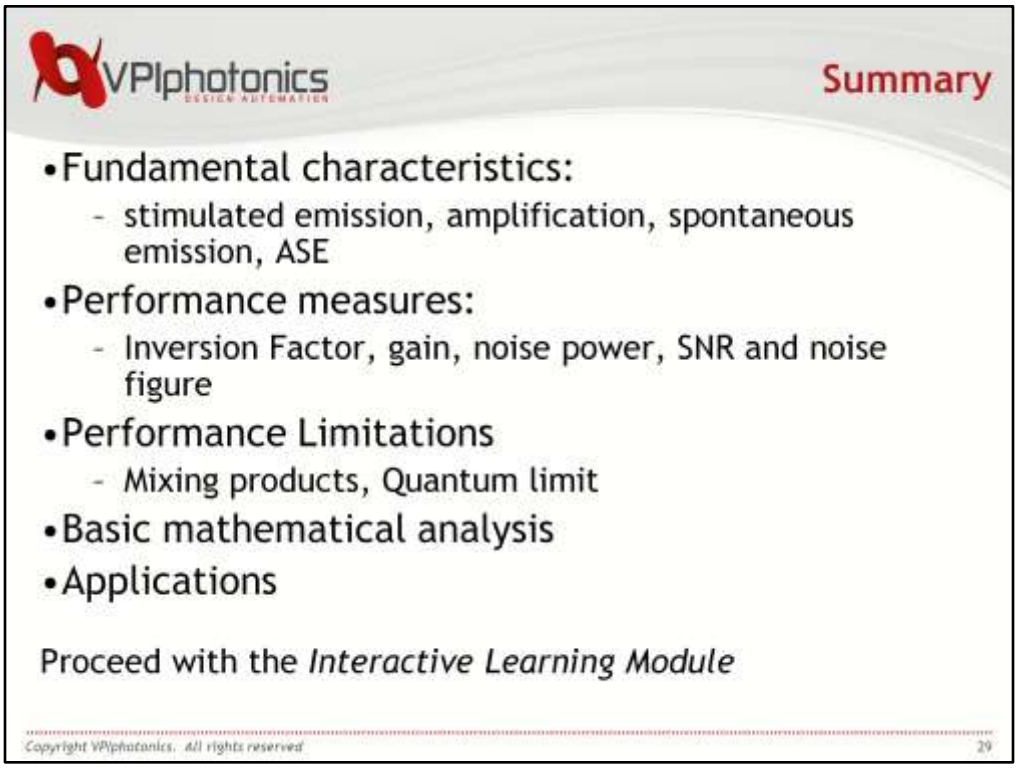
It is possible to, if one tried, to come up with values of G and n_{sp} for which the noise figure is < 2, but they do not represent realisable or practical optical amplifiers (challenge the students to do so, and then to explain why not).



Power amplifier: placed immediately after the laser to boost power for long distance transmission.

Line amplifier: Used to replace repeaters in long-haul systems, compensates for attenuation in fiber spans splitters etc. line amplifies generally require only moderate gain output power and noise performance (typically compensating for 10-20 dB attenuation)

Receiver preamplifier: allows very low level signals to be received. Preamplifiers usually require high gain and very low output noise power.



Summary of the key ideas of the lecture part of this module